performance and accelerations should also be established at
displacement mode during operation in the worst intended condition.
3.2 Operation levels, as defined in 3.1, should be established and documented
by full-scale tests in at least two relevant sea conditions and in head,
beam and following seas. Test period should be at least 15 min. Model
tests and mathematical simulations could be used to verify the
performance in the worst intended conditions.
WI1689n/pp/EWP
RESOLUTION MSC.36(63)
(adopted on 20 May 1994)
ADOPTION OF THE INTERNATIONAL CODE OF SAFETY
FOR HIGH SPEED CRAFT MSC 63/23/Add.2
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Limits for normal operation condition should be documented by
measurements of craft speed, heading to the wave and interpolation of
measurements of maximum horizontal accelerations in accordance with 2.4
of annex 3. Measurement of wave height and period should be made to the
maximum extent practicable.

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SOURCE: MSC.36(63).pdf
measurements of craft speed, heading to the wave and interpolation of
measurements of maximum horizontal accelerations in accordance with 2.4
of annex 3. Measurement of wave height and period should be made to the
maximum extent practicable.
Limits for worst intended condition should be documented by measurements
of craft speed, wave height and period, heading to the wave and by root
mean square (RMS) values of horizontal accelerations in accordance with
2.4 of annex 3 and of vertical accelerations close to the craft
longitudinal centre of gravity. RMS values could be used for
extrapolation of peak values. To obtain the expected peak values related
to structural design load and safety levels (1 per 5-min exceedance),
multiply the RMS values by 3.0 or
C = V2),nN~, -
where:
N is the number of successive amplitudes within the relevant
period.
If not otherwise verified by model tests or by mathematical calculations,
it might be assumed a linear relation between wave height and ·
accelerations based on measurements in the two sea conditions.
for worst intended condition should be documented both related
passenger safety in accordance with 2.4 of annex 3 and related

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SOURCE: MSC.36(63).pdf
accelerations based on measurements in the two sea conditions.
for worst intended condition should be documented both related
passenger safety in accordance with 2.4 of annex 3 and related
actual structural design load of the craft.
Limits
to

to the
3.3 The tests and verification process should document the limiting seas for safe operation of the craft:
.1 in normal operation at maximum operational speed the accelerations should not exceed safety level 1 in annex 3 with an average of 1 per 5-min period. The craft operating manual should include detailed description of the effects of speed reduction or change of heading to the waves in order to prevent exceedance;
.2 in the worst intended conditions, with reduced speed as necessary, the accelerations should not exceed safety level 2 in annex 3 with an average of 1 per 5-min period, nor should any other craft characteristic motion as pitch, roll and yaw exceed levels that could impede the safety of passengers. In worst intended conditions, with reduced speed as necessary, craft should be safely manoeuvrable and provide adequate stability in order that the craft

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SOURCE: MSC.36(63).pdf
could impede the safety of passengers. In worst intended conditions, with reduced speed as necessary, craft should be safely manoeuvrable and provide adequate stability in order that the craft can continue safe operation to the nearest place of refuge, provided caution is exercised in handling. Passengers should be required to be seated when safety level 1 in annex 3 is exceeded; and
.3 within the actual structural design load for the craft, with reduced speed-and change of heading, as necessary.
3.4 Turning and manoeuvrability
The craft should be safely controllable and manoeuvrable during:
.1 hull-borne operation;
.2 operation in non-displacement mode;

.3 take-off, landing;
.4 any intermediate or transition modes, as applicable; and
.5 berthing operations, as applicable.
4 Effects of failures Qr malfunction
4.1 General
The limits of safe operation, special handling procedures and any operational restrictions should be examined and developed as a result of

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SOURCE: MSC.36(63).pdf
.5 berthing operations, as applicable.
4 Effects of failures Qr malfunction
4.1 General
The limits of safe operation, special handling procedures and any operational restrictions should be examined and developed as a result of

full-scale trials conducted by simulating possible equipment failures. The failures to be examined should be those leading to major or · more severe effects as determined from evaluation of FMEA or similar analysis. Failures to be examined should be agreed between the craft manufacturer and the Administration and each single failure should be examined in a progressive manner.

4.2 Objects of tests

Examination of each failure should result in:

.1 determining safe limits of craft operation at the time of failure, beyond which the failure will result in degradation beyond safety level 2;

.2 determining crew member's actions, if any, to minimize or counter the effect of the failure; and

.3 determining craft or machinery restrictions to be observed to enable the craft to proceed to a place of refuge with the failure present.

4.3 Failures to be examined

Equipment failures should include, but not be limited to, the following:

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SOURCE: MSC.36(63).pdf

.3 determining craft or machinery restrictions to be observed to enable the craft to proceed to a place of refuge with the failure present.

4.3 Failures to be examined

Equipment failures should include, but not be limited to, the following:

.1 total loss of propulsion power;

.2 total loss of lift power (for ACV and SES);

.3 total failure of control of one propulsion system;

.4 involuntary application of full propulsion thrust (positive or negative) on one system;

.5 failure of control of one directional control system;

.6 involuntary full deflection of one directional control system;

.7 failure of control of trim control system;

.8 involuntary full deflection of one trim control system element; and

.9 total loss of electrical power.

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Failures should be fully representative of service conditions and should be simulated as accurately as possible in the most critical craft manoeuvre where the failul'e will have maximum impact.

4.4 "Dead ship" test

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Failures should be fully representative of service conditions and should be simulated as accurately as possible in the most critical craft manoeuvre where the failul'e will have maximum impact.

4.4 "Dead ship" test

In order to establish craft motions and direction of laying to wind and waves, for the purposes of determining the conditions of a craft evacuation, the craft should be stopped and all main machinery shut down for sufficient time that the craft's heading relative to wind and waves has stabilized. This test should be carried out on an opportunity basis to establish patterns of the design's "dead ship" behaviour under a variety of wind and sea states.

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ANNEX 9

CRITERIA FOR TESTING AND EVALUATION OF
REVENUE AND CREW SEATS

1 Purpose and scope